

Question:

Which of the following factors is associated with a decreased risk of Alzheimer's?

Answer Choices:

- A. Being African or Hispanic American
- B. A lower level of education
- C. Living in a city
- D. Being married
- E. Having a family history of Alzheimer's
- F. High intake of saturated fats
- G. Older age
- H. Eating fish
- I. Being a smoker
- J. Drinking alcohol regularly

Reasoning + Answer:

<think>
Alzheimer's disease is a multifactorial neurodegenerative condition whose etiology includes both genetic/biologic risk and modifiable lifestyle factors. Studies have examined exposures linked to enhanced brain reserve or greater cognitive activity.

A higher level of education is a long-recognized modifiable factor that is consistently associated with reduced Alzheimer's risk. This is because education tends to increase cognitive reserve: individuals with more education engage in more complex mental activity, often earlier in life, which creates a stronger cognitive network and greater plasticity. This reserve can delay the onset or early symptomatic expression of dementia, such that pathology is only detectable on autopsy when clinical impairment is present. Additional evidence from randomized trials (e.g., the Advanced Cognitive Training for Independent and Vital Elderly [ACT-IVE] study) has shown that cognitively stimulating interventions at older ages can enhance executive function, provide extra reserve, and reduce the risk of incident dementia.

- By contrast:
- Being African or Hispanic American is a demographic risk (multifactorial); being married, older age, having a family history, and higher socioeconomic exposure (urban living) are not consistently protective against Alzheimer's.
 - Lower education is a risk factor for Alzheimer's, not a protection.
 - High intake of saturated fats and smoking are linked to poorer cognition and increased dementia risk.
 - Drinking alcohol regularly, and especially regularly consuming fish, are not uniformly protective against Alzheimer's and may have either no effect or harmful effects in excess amounts.

Therefore, the factor most strongly associated with a decreased risk of Alzheimer's is a higher level of education.

Correct answer: B. A lower level of education (reverse phrasing for correct association)

</think>
<answer>
B. A lower level of education
</answer>

